

Motivation

Modeling action-induced non-stationarity phenomena in online learning:

- Deprivation: Payoff of an action increases after an interval of inactivity.
- Satiation: Consecutively playing the same action decreases its payoff.

Applications: News-feeds, recommendation systems, ad-placement etc.

Related Work

Recharging Bandits

- Payoff of each arm i is a non-decreasing function $p_i(\cdot)$ of the rounds passed since it was last played. At most k arms can be pulled per round.
- **Immorlica and Kleinberg** [2]: $(1 - \epsilon)$ -PTAS for the planning problem for $k = 1$, when $p_i(\cdot)$ is also weakly concave and Lipschitz.
- **Simchi-Levi et al.** [6]:

$$\max_{a \in \mathbb{Z}_+} \frac{a}{a+1} \cdot \frac{k}{k+a} \text{-approximation for the planning problem for any } k.$$

- **Papadigenopoulos et al.** [5]:

$$\left(1 - \frac{k^k}{e^k k!}\right) \text{-approximation for the planning problem for any } k.$$

- The above can be adapted to the bandit setting with sublinear regret.

Blocking Bandits

- Payoff of each arm becomes zero for a number of rounds after each play.
- Special case of Recharging Bandits when $p_i(\cdot)$ are translated Heaviside step functions.
- **Basu et al.** [1] and **P. & Caramanis** [4]: $1 - \frac{1}{e}$ -approximation for the planning problem for any k and sublinear regret for the bandit.

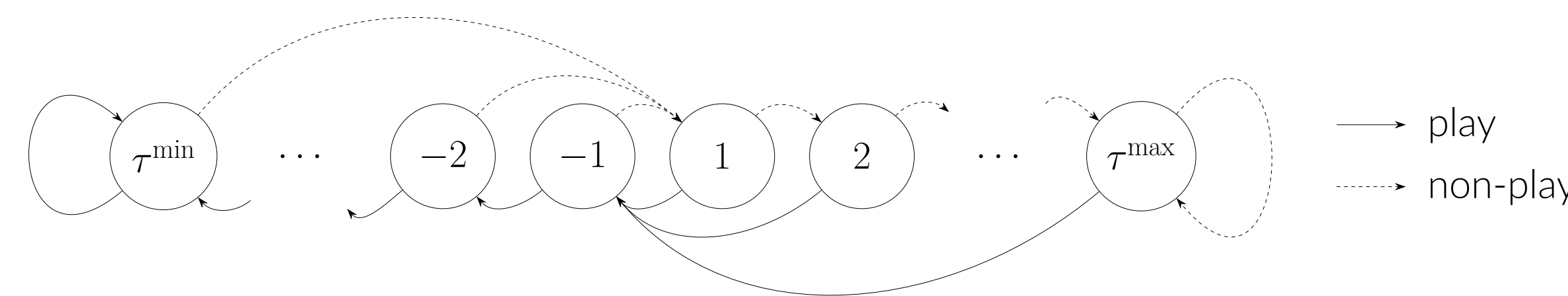
Last Switch Dependent Bandits

- Introduced by **Laforgue et al.** [3] (see next)
- Generalization of Recharging Bandits capturing satiation and seasonality.
- Payoff functions are monotone only over positive states $\tau \geq 1$.
- Branch-and-bound based approach and reduction to Combinatorial Semi-Bandits.

LSD Bandits with Monotone Payoff functions

Model

- Set of n arms (actions) and unknown time horizon T .
- Each arm i has an internal state $\tau_i \in \mathbb{Z} \setminus \{0\}$ that changes according to the following transition graph,



- Each arm i is associated with an unknown **monotone** mean payoff function $p_i(\cdot) : \mathbb{Z} \setminus \{0\} \rightarrow [0, 1]$ such that $p_i(\tau)$ is the mean payoff of arm i when played under state τ .

Objective

- Player can pull at most $k < n$ arms in each round.
- Observes the payoff of each arm played and collects their sum.
- **Planning objective (known payoff functions):** Maximize the total expected payoff collected over T rounds (NP-Hard problem).
- **Bandit objective (unknown payoff functions):** For a given approximation factor γ for planning, minimize the γ -regret,

$$\text{Reg}(T) = \gamma \text{OPT}(T) - R(T),$$

where $\text{OPT}(T)$ is the optimal total expected payoff and $R(T)$ is the total expected payoff collected by the bandit algorithm.

Main Results

Planning Setting

We propose a randomized algorithm for the planning problem that is a

$$(1 - \epsilon) \left(1 - \frac{k^k}{e^k k!}\right) \underset{k \rightarrow \infty}{\sim} (1 - \epsilon) \left(1 - \frac{1}{\sqrt{2\pi k}}\right) \text{-approximation}$$

for any fixed $\epsilon > 0$ in time polynomial in $1/\epsilon$ and other parameters of the problem.

Algorithm

LP relaxation

- We introduce the notion of a **recurrent interval** (a minimal sequence of plays and non-plays that can be played cyclically).
- We give an **LP relaxation** whose variables represent the fraction of time an arm is played under a particular recurrent interval.

Constructing a virtual evolution for each arm

- We sample a unique recurrent interval for each arm using the optimal solution to our LP relaxation.
- Periodically traversing over each sampled recurrent interval gives a “**virtual**” evolution of the state of each arm.

Mirroring the virtual evolutions

- Ideally, implementing exactly the virtual evolution of all arms would recover the LP solution, but it is generally infeasible.
- We approximately “**mirror**” the virtual evolutions by randomly interleaving them and playing, at each round, the k arms of maximum virtual payoff.

Bandit Setting

We adapt our planning algorithm to the bandit setting and achieve a regret bound of $\tilde{O}\left(n^{\frac{1}{3}} k^{\frac{2}{3}} T^{\frac{2}{3}}\right)$.

References

- [1] Soumya Basu, Rajat Sen, Sujay Sanghavi, and Sanjay Shakkottai. Blocking bandits. *Advances in Neural Information Processing Systems*, 2019.
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- [3] Pierre Laforgue, Giulia Clerici, Nicolò Cesa-Bianchi, and Ran Gilad-Bachrach. A last switch dependent analysis of satiation and seasonality in bandits. In *International Conference on Artificial Intelligence and Statistics*. PMLR, 2022.
- [4] Orestis Papadigenopoulos and Constantine Caramanis. Recurrent submodular welfare and matroid blocking semi-bandits. *Advances in Neural Information Processing Systems*, 2021.
- [5] Orestis Papadigenopoulos, Constantine Caramanis, and Sanjay Shakkottai. Non-stationary bandits under recharging payoffs: Improved planning with sublinear regret. In *Advances in Neural Information Processing Systems*, 2022.
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